# Better Executives Work at Better Firms: Identifying Talent Sorting from Mobility Networks

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# Motivation

# Does the allocation of executive talent matter for GDP?

- Misallocation lowers aggregate productivity (Hsieh and Klenow 2009)
- Talent is a fixed factor: who runs which firm matters
- Executives are few, but their decisions scale with firm size
- This is a macro question: we need the *whole* economy, not a selected sample

# A simple assignment model

Firm $i$ has fundamentals $A_i$, executive $m$ has talent $Z_m$. With variable inputs substituted out, revenue is Cobb–Douglas in the two *fixed factors*:

$$Y_{im} = A_i^{1-\nu} Z_m^{\nu}$$

## Two implications

- Diminishing returns to each factor pin down firm scale
- $\partial^2 Y / \partial A \, \partial Z > 0$: firm fundamentals and talent are **complements**

## Complementarity $\Rightarrow$ assignment matters

The efficient allocation matches better executives to better firms (Becker 1973, Lucas 1978)

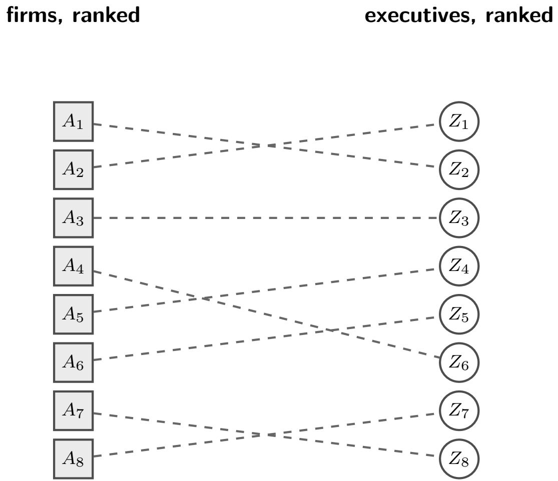
# Sorting shows up directly in GDP

Sum output across the $n$ matches in the economy. With $a = \ln A$, $z = \ln Z$ jointly normal (law of large numbers),

$$\frac{Y}{n} = \exp\left\{(1-\nu)\mu_a + \nu\mu_z + \tfrac{1}{2}\left[(1-\nu)^2\sigma_a^2 + \nu^2\sigma_z^2\right] + \nu(1-\nu)\,\mathrm{Cov}(a,z)\right\}$$

- Marginal distributions fixed: GDP is increasing in $\mathrm{Cov}(a,z)$
- **The covariance of talent and fundamentals across matches is an aggregate outcome**
- How large is it in the data?

# Talent allocation in one picture




Sorting is positive but imperfect. How far is the economy from the diagonal?

# The question and the challenge

## Layer 1: a macro question

How strongly are productive firms matched with talented executives, and what does the allocation contribute to aggregate output?

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How strongly are productive firms matched with talented executives, and what does the allocation contribute to aggregate output?

## Layer 2: measuring something with nothing

Neither firm fundamentals $A_i$ nor executive talent $Z_m$ is observable. How do you estimate a correlation between two latent variables?

# How can we measure executive talent?

- Surveys and diary studies of what executives do (Bandiera et al 2020)
- Observable characteristics: education, experience, style
- **From outcomes:** executives who move reveal their persistent contribution (Bertrand and Schoar 2003)
- Matched two-way decompositions in labor (Abowd, Kramarz and Margolis 1999)

We follow the outcome-based route — but with a twist.

# What we do

1. Use the **executive mobility network**: chains of moves connect firms and executives that never meet
2. We do *not* estimate a latent talent for every executive
3. We estimate the parameters of a **sorting model**, in particular $\mathrm{Corr}(a, z)$

# Our object of interest: four parameters

Model $(A_i, Z_m)$ of matched pairs as jointly lognormal, add match noise:

$$y_{im} = a_i + z_m + \varepsilon_{im}, \qquad \begin{pmatrix} a_i \\ z_m \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \sigma_a^2 & \rho\sigma_a\sigma_z \\ \rho\sigma_a\sigma_z & \sigma_z^2 \end{bmatrix}\right)$$

## Four parameters

$$\sigma_a, \quad \sigma_z, \quad \rho, \quad \sigma_\varepsilon$$

- $\rho =$ sorting: the correlation between firm fundamentals and executive talent
- We estimate these on the full economy, using the network
- How? Hold that thought — first, what everyone else does

# Prior art: fixed effects

# Mobility reveals talent

- A firm switches executives; performance jumps
- An executive improves *every* firm they run
- Statistical evidence that their talent exceeds their predecessor's

## The fixed-effects route

Take logs of the lognormal model and treat $a_i$, $z_m$ as *parameters*:

$$y_{im} = a_i + z_m + \varepsilon_{im} \quad \Rightarrow \quad \text{OLS with two sets of dummies}$$

- Labor: (Abowd, Kramarz and Margolis 1999; Andrews et al 2008; Kline, Saggio and Sølvsten 2020)
- Executives and managers: (Bertrand and Schoar 2003; Fenizia 2022; Metcalfe, Sollaci and Syverson 2023)

# Two problems with fixed effects

## 1. Only works on a connected component

Effects are identified only *within* a connected component of the mobility graph. Standard practice: keep the largest one, drop the rest.

## 2. Noise makes second moments biased

Because of $\varepsilon$, estimated effects are noisy. Second moments — exactly what we care about — are biased: $\widehat{\mathrm{Var}}$ too big, $\widehat{\mathrm{Cov}}$ too small (Andrews et al 2008; Koren, Orbán and Telegdy 2025)

Limited mobility makes both problems worse.

# Both problems are worse for executives

- Executive mobility graph is **very sparse**: few moves per person
- Hundreds of thousands of disconnected components
- Firm outcomes are noisy
- Estimates especially noisy — *and* the usual corrections do not apply

Why corrections fail: next slide.

# The leave-one-out correction and leverage

Idea (Kline, Saggio and Sølvsten 2020): estimate each effect *without* observation $i$, so noise does not contaminate its own estimate.

- FE estimates are linear in outcomes; observation $i$ enters its own fitted value with weight $h_i$ = **leverage**
- Bias correction re-weights by $1/(1 - h_i)$

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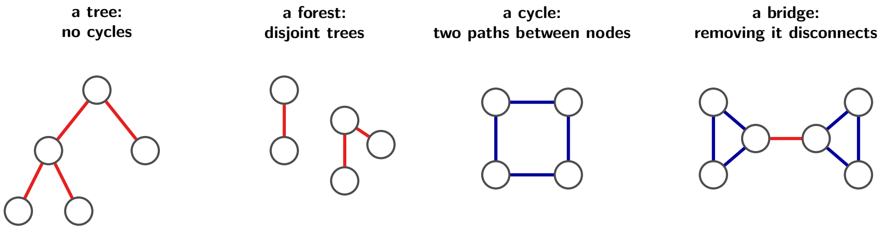
## The catch

$h_i = 1$: the effect is estimable *only* from observation $i$. Dropping it, we know nothing — the correction divides by zero.

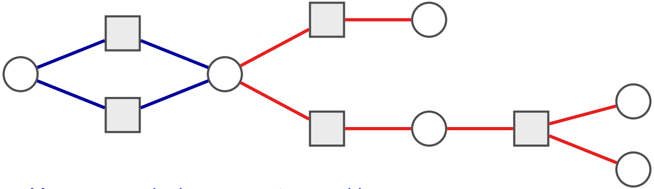
$h_i = 1$ **exactly when edge $i$ is a bridge.**

# A detour: trees, forests, cycles, bridges




- **Bridge** (red): the only path between its two sides — leverage $= 1$
- On a tree, *every* edge is a bridge
- Bridges also connect cycles: tree-like edges in a general graph

# Which edges can the leave-out correction use?



**blue**: on a cycle, leverage $< 1$ — usable

**red**: bridges, leverage $= 1$ — dropped by leave-out

Squares are firms, circles are executives, edges are jobs. Leave-out keeps only edges on cycles.

# The Hungarian executive mobility network

Universe of Hungarian corporations and their registered chief executives, 1990–2022. An *observation* is an edge with an outcome: firm performance under that executive.

| | Value |
|---|---|
| Executives | 1,304,580 |
| Firms | 1,036,231 |
| Firm–executive edges | 1,937,552 |
| Connected components | 514,085 |
| **Bridge share of edges** | **81.8%** |
| Giant component: share of executives | 29.6% |
| Giant component: share of firms | 33.0% |
| Mean shortest path between executives (giant) | 19.5 |

The graph is close to a forest, with long, sparse chains.

# What is left after the standard corrections?

| Stage | Matched pairs | % |
|---|---|---|
| All unique firm–executive pairs | 1,732,154 | 100.0% |
| Drop bridges (leverage $= 1$) | 265,995 | 15.4% |
| Keep largest leave-out-connected block | 175,661 | **10.1%** |

- Leave-out estimation keeps **one in ten** matched pairs
- Survivors are highly mobile executives at highly connected firms — not representative
- For a macro question about the whole economy, this will not do

# Our approach: a random-effects model on the network

# We model the joint distribution of all outcomes

Instead of estimating 2.3 million fixed effects, we model the **joint distribution** of all 1.7 million outcomes on the network:

$$\mathbf{x} = (a_1, \ldots, a_{N_f}, z_1, \ldots, z_{N_m})^\top \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^{-1}), \qquad y_{im} = a_i + z_m + \varepsilon_{im}$$

- $\mathbf{Q}$: sparse **precision matrix** encoding the mobility graph
- A **Gaussian Markov random field** (Rue and Held 2005)
- Same four parameters as before: $\sigma_a$, $\sigma_z$, $\rho$, $\sigma_\varepsilon$

# The precision matrix, one edge at a time

$$\mathbf{Q} = \frac{1}{1-\rho^2}\,\mathbf{S}\left[(1-\rho^2)\mathbf{I} + \rho^2\mathbf{D} - \rho\mathbf{A}\right]\mathbf{S}, \qquad \mathbf{S} = \mathrm{diag}(\sigma_a^{-1}\mathbf{I}_{N_f},\, \sigma_z^{-1}\mathbf{I}_{N_m})$$

$\mathbf{A}$: adjacency, $\mathbf{D}$: degrees. Zoom in on a single firm–executive pair (degree 1 each):

$$\mathbf{Q}_{\{i,m\}} = \frac{1}{1-\rho^2}\begin{bmatrix} \sigma_a^{-2} & -\rho\,\sigma_a^{-1}\sigma_z^{-1} \\ -\rho\,\sigma_a^{-1}\sigma_z^{-1} & \sigma_z^{-2} \end{bmatrix} \iff \mathrm{Var}\begin{pmatrix} a_i \\ z_m \end{pmatrix} = \begin{bmatrix} \sigma_a^2 & \rho\,\sigma_a\sigma_z \\ \rho\,\sigma_a\sigma_z & \sigma_z^2 \end{bmatrix}$$

- Off-diagonal entries of $\mathbf{Q}$ are nonzero **only along observed edges**
- On a forest, every linked pair has correlation exactly $\rho$, variances $\sigma_a^2$, $\sigma_z^2$

# Detour: Gaussian Markov random fields

# What is a Gaussian Markov random field?

A way to model jointly normal variables with a **sparse precision matrix**.

- Zero in $\mathbf{Q}$ $\Leftrightarrow$ conditional independence given everyone else
- *Conditional* dependence is local: only direct neighbors on the graph
- *Unconditional* dependence can reach far — it travels along paths

The graph disciplines 2.3M-dimensional $\mathbf{x}$ with a handful of parameters.

# The best-known GMRF: an AR(1)

$x_t = \rho x_{t-1} + u_t$ on a line graph. Its precision matrix is tridiagonal:

$$\mathbf{Q} \propto \begin{bmatrix} 1 & -\rho & & \\ -\rho & 1+\rho^2 & -\rho & \\ & -\rho & 1+\rho^2 & -\rho \\ & & -\rho & 1 \end{bmatrix}$$

- Partial correlation between neighbors: governed by $\rho$
- Unconditional autocorrelation at lag $d$: $\rho^d$ — decays with distance
- Sparse precision, dense covariance

# On a tree, the GMRF looks just like an AR(1)

Claim: on a tree, the unconditional covariance between any two nodes at graph distance $d$ is

$$\mathrm{Cov}(x_k, x_\ell) = \sigma_k \sigma_\ell \, \rho^{\,d(k,\ell)}$$

- Exactly the AR(1) pattern — the tree is a branching timeline
- Unique paths $\Rightarrow$ correlation decays geometrically per hop
- Our variance-stable $\mathbf{Q}$ keeps marginals at $\sigma_a^2$, $\sigma_z^2$ on any forest

# End of detour: real networks have cycles

- Cycles create multiple paths $\Rightarrow$ covariances above $\rho^d$
- The GMRF handles this exactly: all paths enter through $\mathbf{Q}^{-1}$
- We estimate by **maximum likelihood** (computation: a section of its own)
- And recall: our graph is 81.8% bridges — *almost* a forest, so tree intuition nearly exact

# Identification: covariance decay along the network

# Covariance decays along the tree

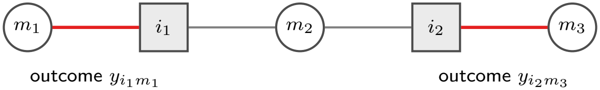
For two outcomes whose executives are $h$ edges apart on a tree:

$$\mathrm{Cov}(y_{i_1 m_1}, y_{i_2 m_2}) \;=\; \rho^{\,h-2}\,(\sigma_a + \rho\sigma_z)^2$$

- **Level** of covariance: mixes $\sigma_a$, $\sigma_z$, $\rho$
- **Speed of decay**: two extra hops multiply covariance by $\rho^2$ — *nothing else*
- Four parameters, many covariance moments: over-identified
- Variance moments also load on $\sigma_\varepsilon^2$; covariances between distinct matches do not

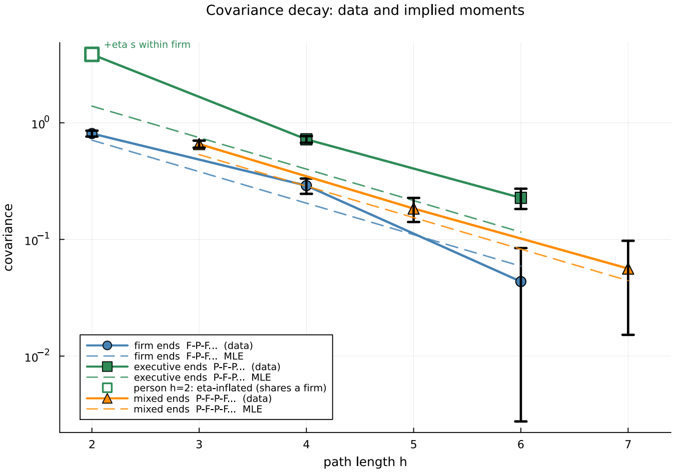
Inspired by correlation decay across generations of family networks (Clark 2014, Clark 2023)

# Prima facie evidence of sorting: the 3-hop covariance




- $m_1$ and $m_3$ never work at the same firm; $i_1$ and $i_2$ never share an executive
- **No common shocks** — yet their outcomes are correlated in the data
- The model says exactly how much: $\rho^2(\sigma_a + \rho\sigma_z)^2$
- Ratios of such moments deliver $\rho^2$ by method of moments — before any likelihood

# Covariance decay in the Hungarian data



# Placebo: disconnected components

- Model prediction: outcomes in *disjoint* components have **zero** covariance
- Any story of common industry or regional shocks predicts otherwise

# What about within-firm shocks over time?

Concern: consecutive spells at the *same firm* share transitory conditions — correlated $\varepsilon$, not sorting.

## Why this is not our identification

- Common shocks are **localized**: they move nearby covariances
- Sorting is identified by the **speed of decay over long chains**

## Three fixes that agree

1. Minimum distance: drop the contaminated same-firm moment
2. Firm-average estimator: allow arbitrary within-firm dependence
3. AR(1) match shocks within firm: estimate the error process ($5$th parameter $\eta$)

# Estimation and computation

# Maximum likelihood at scale

$$\ell(\boldsymbol{\theta}) = -\frac{K}{2}\ln(2\pi) - \frac{1}{2}\ln\det\boldsymbol{\Omega} - \frac{1}{2}\mathbf{y}^\top\boldsymbol{\Omega}^{-1}\mathbf{y}, \qquad \boldsymbol{\Omega} = \mathbf{V}\mathbf{Q}^{-1}\mathbf{V}^\top + \sigma_\varepsilon^2\mathbf{R}_\eta$$

- $\boldsymbol{\Omega}$ is $1.7$M $\times$ $1.7$M and dense — direct inversion infeasible
- Woodbury: work with sparse $\mathbf{P} = \mathbf{Q} + \mathbf{V}^\top(\sigma_\varepsilon^2\mathbf{R}_\eta)^{-1}\mathbf{V}$ in node space
- Exact sparse Cholesky gives log-determinant and quadratic form
- 2.3M latent nodes; converges on a single machine (Julia)

# Keeping the model well-defined: pruning

- $\mathbf{Q} \succ 0$ requires $|\rho|\,\lambda_{NB} < 1$: a ceiling set by the graph's cycle structure (non-backtracking spectral radius)
- We prune a few cycle-closing edges in one dense region — keeping *every* node and component
- Resulting ceiling: $\rho < 0.666$; estimates are interior
- The graph is fixed before seeing outcomes

# Results

# Sorting is high: $\rho \approx 0.52$–$0.55$

All estimators on the same graph and outcomes (log real revenue, demeaned by year and industry):

| Parameter | Naive MLE | Min. distance | Firm-average | AR(1) MLE |
|---|---|---|---|---|
| $\rho$ | 0.379 | 0.555 | 0.537 | **0.518** |
| Implied corr. $(a, z)$ | 0.381 | 0.555 | 0.544 | 0.525 |

- Naive i.i.d.-error MLE is biased *down*: within-firm shocks masquerade as firm heterogeneity
- Three different corrections for within-firm dependence **agree**: $0.52$–$0.55$
- Preferred AR(1) fit: $\hat{\sigma}_a = 1.08$, $\hat{\sigma}_z = 0.24$, $\hat{\sigma}_\varepsilon = 1.72$, $\hat{\eta} = 0.51$

# Where does revenue variance come from?

At the preferred fit, share of log-revenue variance:

| Component | Share |
|---|---|
| Firm effects | 28.4% |
| Executive effects | 1.4% |
| Sorting covariance | 6.6% |
| Match noise (within-firm correlated) | 63.5% |

- Sorting covariance is **several times** the executive component
- Executives matter mostly through *where they are allocated*

# Random assignment would cost 12.9% of revenue

Log-normal accounting benchmark, marginals held fixed:

$$\frac{Y}{n} = \exp\left\{\mu_a + \mu_z + \mu_\varepsilon + \tfrac{1}{2}(V_a + V_z + V_\varepsilon + V_{\text{cross}})\right\}, \qquad V_{\text{cross}} = 2\,\mathrm{Cov}(a, z)$$

- Random assignment sets $V_{\text{cross}} = 0.275 \to 0$
- Mean revenue falls by 0.138 log units = **12.9%**
- Allocation of the observed sector, not an economy-wide GDP counterfactual

# Conclusion

1. Sorting between executive talent and firm fundamentals is **high**: $\rho \approx 0.52$–$0.55$
2. A four-parameter random-effects model replaces millions of fixed effects — and uses **all** the data, not 10%
3. Identification: covariance decay along mobility chains — transparent, testable
4. Allocation matters: random assignment would lower revenue by 12.9%

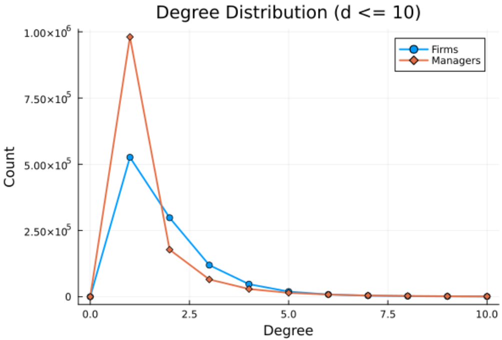
## Method travels

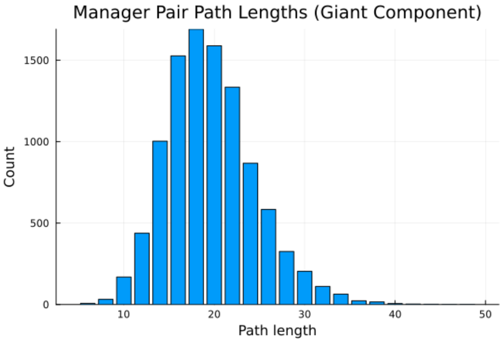
Any sparse bipartite matching network: workers–firms, students–schools, patients–providers

# Backup slides

# The bipartite mobility graph

- Nodes: firms (squares) and executives (circles); edges: unique firm–executive links
- 514,085 components; the giant holds 42% of edges, the rest mostly single pairs
- Degrees are tiny; paths are long

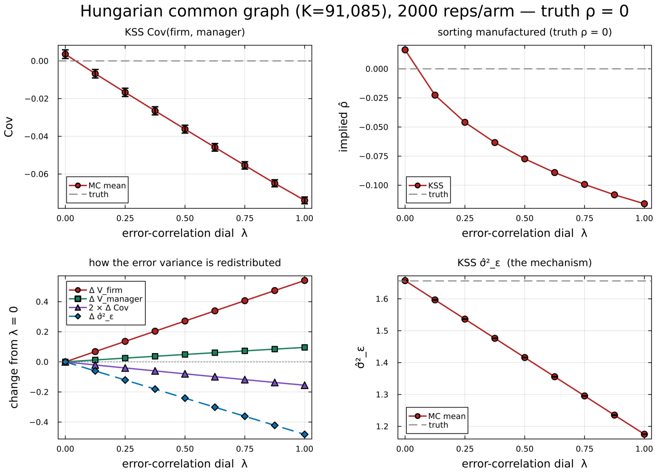


# Multi-executive spells

- A match cluster: firm interval with a constant set of co-present executives
- One outcome, one error term; design row loads $1/q$ on each of $q$ executives
- Avoids crediting every executive with the full firm outcome

# Monte Carlo: correlated errors on the Hungarian graph



# Feasibility ceiling and pruning diagnostics